

TASK 1 : DNA/Protein Sequence Analysis and BLAST Homology Search

1. INTRODUCTION :

Bioinformatics provides computational approaches for analyzing biological sequences and identifying relationships between proteins and genes. Sequence similarity analysis is an important technique for understanding the possible function, evolutionary conservation and relationships of biological molecules. BLAST (Basic Local Alignment Search Tool) is widely used to compare a query sequence against biological sequence databases and identify similar or homologous sequences.

2. OBJECTIVE :

The objective of this study was to retrieve the human TP53 protein sequence from UniProt and perform a BLASTP search against a protein sequence database to identify homologous sequences and evaluate their sequence similarity.

3. PROTEIN INFORMATION

Parameter	Information
Protein	Cellular tumor antigen p53
Gene	TP53
Organism	Homo sapiens
UniProt Accession	P04637
Sequence Type	Protein
BLAST Program	BLASTP
Database	nr

4. SEQUENCE RETRIEVAL :

The human TP53 protein entry was retrieved from the UniProt database using accession P04637. The protein sequence was obtained in FASTA format and subsequently used as the query sequence for BLASTP analysis.

FIG 1: UniProt entry and sequence information for human TP53 protein.

Bioinformatics Tasks & Instructions | human TP53 in UniProtKB search | + | Ask Gemini

uniprot.org/uniprotkb?query=human+TP53

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB human TP53 Advanced | List Search Help

Status

- Reviewed (Swiss-Prot) (1,149)
- Unreviewed (TrEMBL) (5,395)

Popular organisms

- Human (1,460)
- Mouse (128)
- Zebrafish (50)
- Rat (25)
- Fruit fly (21)

UniProtKB 6,544 results

Unique and stable entry identifier. d (7k) Add View: Cards Table Customize columns Share

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
P04637	P53_HUMAN	Cellular tumor antigen p53[...]	TP53, P53	Homo sapiens (Human)	393 AA
Q9Y2B4	T53G5_HUMAN	TP53-target gene 5 protein[...]	TP53TG5, C20orf10	Homo sapiens (Human)	290 AA
O95905	ECD_HUMAN	Protein ecdysoneless homolog[...]	ECD	Homo sapiens (Human)	644 AA
Q12888	TP53B_HUMAN	TP53-binding protein 1[...]	TP53BP1	Homo sapiens (Human)	1,972 AA

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uniprot.org/uniprotkb/P04637/entry

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|Function

Names & Taxonomy

Subcellular Location

Disease & Variants

PTM/Processing

Expression

Interaction

Structure

Family & Domains

Sequence & Isoforms

Similar Proteins

Homologs

P04637 - P53_HUMAN

Protein: Cellular tumor antigen p53

Gene: TP53

Status: UniProtKB reviewed (Swiss-Prot)

Organism: Homo sapiens (Human)

Amino acids: 393 (go to sequence)

Protein existence: Evidence at protein level

Annotation score: 5/5

Tools Download Add Community curated (2) Add a publication Entry feedback

Entry Variant viewer 3,551 Feature viewer Genomic coordinates Publications External links History

Function

Multifunctional transcription factor that induces cell cycle arrest, DNA repair or apoptosis upon binding to its target DNA sequence (PubMed:11025664, PubMed:12524540, PubMed:12810724, PubMed:15186775, PubMed:15340061, PubMed:17317671, PubMed:17349958, PubMed:19556538, PubMed:20673990, PubMed:20959462, PubMed:22726440, PubMed:24051492, PubMed:24652652, PubMed:35618207, PubMed:36634798, PubMed:38653238, PubMed:9840937). Acts as a tumor suppressor in many tumor types; induces growth arrest or apoptosis depending on the physiological circumstances and cell type

Help

5. BLAST ANALYSIS :

The TP53 protein sequence was submitted to the NCBI BLASTP tool to identify homologous protein sequences. The non-redundant protein sequence database was used for sequence comparison. The BLAST search generated multiple hits that were ranked according to alignment scores and statistical significance.

FIG 2: Submission of the TP53 protein sequence to NCBI BLASTP.

Bioinformatics Tasks & Instructions | rest.uniprot.org/uniprotkb/P04637 | Protein BLAST: search protein | + | Ask Gemini

blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins

National Library of Medicine National Center for Biotechnology Information

BLAST® = blastp suite

Standard Protein BLAST

blastn blastx tblastn tblastx

Enter Query Sequence

Enter accession numbers, gis, or FASTA sequences

Query subrange

From To

OR against the

Choose File No file chosen

Job Title

Enter a descriptive title for your BLAST search

Align two or more sequences

Choose Search Set

Database

Organism

Program Selection

Algorithm

BLAST

Search database ClustalW using Blastp (protein-protein BLAST)

Show results in a new window

+ Algorithm parameters

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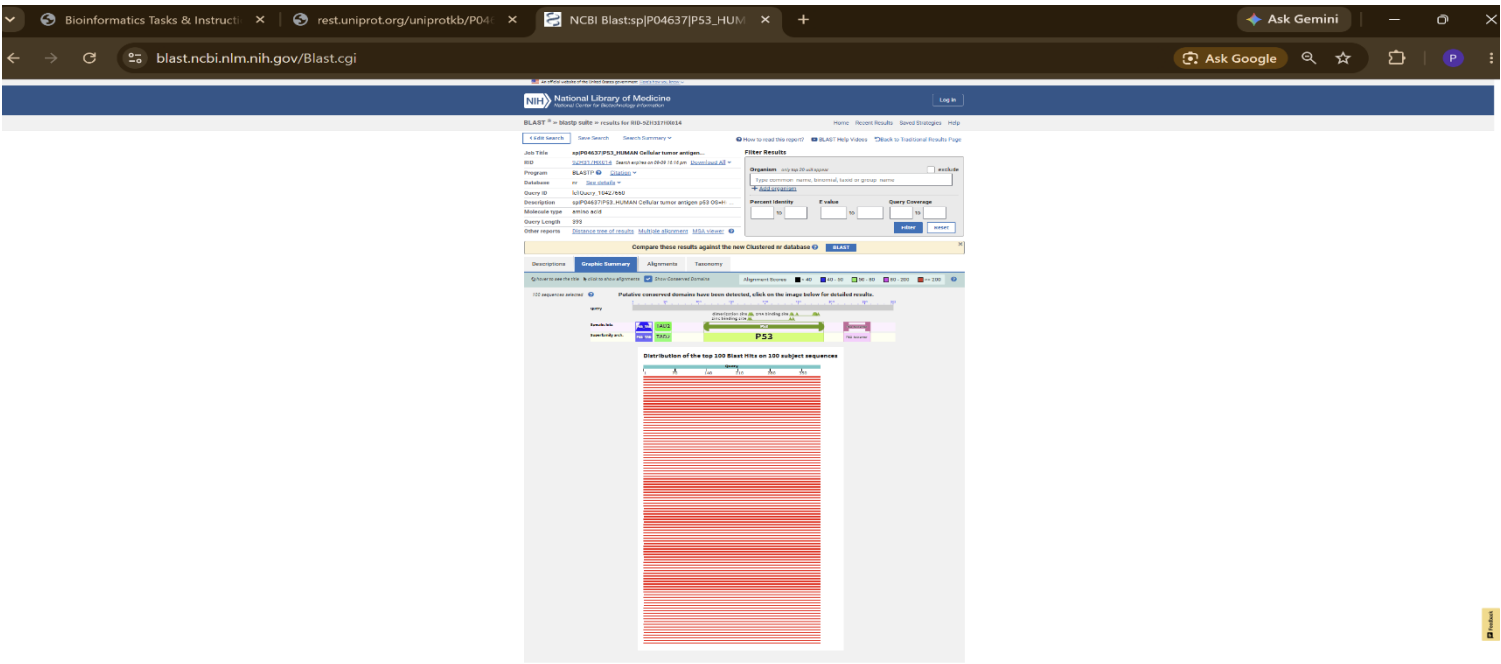
FOIA

1816 Vulnerability Disclosure

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6. BLAST RESULTS :



7. TOP BLAST HITS :

An official website of the United States government																																																														
National Library of Medicine National Center for Biotechnology Information					Log in																																																									
BLAST [®] » blastp suite » results for RID-9ZH317HX014					Home Recent Results Saved Strategies Help																																																									
◀ Edit Search Save Search Search Summary					How to read this report? BLAST Help Videos Back to Traditional Results Page																																																									
Job Title spP04637IP53_HUMAN Cellular tumor antigen... RID 9ZH317HX014 Search expires on 09-09 16:16 pm Download All Program BLASTP Citation Database nr See details Query ID IcdQuery_10427660 Description spP04637IP53_HUMAN Cellular tumor antigen p53 OS=H... Molecule type amino acid Query Length 393 Other reports Distance tree of results Multiple alignment MSA viewer					Filter Results Organism only top 20 will appear Type common name, binomial, taxid or group name Add organism Percent Identity E value Query Coverage Filter Reset																																																									
Compare these results against the new Clustered nr database BLAST																																																														
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Sequences producing significant alignments Download Select columns Show 100																																																														
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<table><thead><tr><th>Description</th><th>Scientific Name</th><th>Max Score</th><th>Total Score</th><th>Query Cover</th><th>E value</th><th>Per Ident</th><th>Acc. Len</th><th>Accession</th></tr></thead><tbody><tr><td>cellular tumor antigen p53 isoform X1 (Gorilla gorilla gorilla)</td><td>Gorilla gorilla gorilla</td><td>814</td><td>814</td><td>100%</td><td>0.0</td><td>99.75%</td><td>408</td><td>XP_003558859.1</td></tr><tr><td>Chain K, Cellular tumor antigen p53 (Homo sapiens)</td><td>Homo sapiens</td><td>814</td><td>814</td><td>100%</td><td>0.0</td><td>100.00%</td><td>396</td><td>TX22_K</td></tr><tr><td>Chain C, Cellular tumor antigen p53 (Homo sapiens)</td><td>Homo sapiens</td><td>814</td><td>814</td><td>100%</td><td>0.0</td><td>100.00%</td><td>417</td><td>8P1F_C</td></tr><tr><td>cellular tumor antigen p53 isoform a (Homo sapiens)</td><td>Homo sapiens</td><td>813</td><td>813</td><td>100%</td><td>0.0</td><td>100.00%</td><td>393</td><td>NP_000837.3</td></tr><tr><td>mutant tumor protein p53 (Homo sapiens)</td><td>Homo sapiens</td><td>813</td><td>813</td><td>100%</td><td>0.0</td><td>99.75%</td><td>393</td><td>AYE20917.1</td></tr></tbody></table>									Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession	cellular tumor antigen p53 isoform X1 (Gorilla gorilla gorilla)	Gorilla gorilla gorilla	814	814	100%	0.0	99.75%	408	XP_003558859.1	Chain K, Cellular tumor antigen p53 (Homo sapiens)	Homo sapiens	814	814	100%	0.0	100.00%	396	TX22_K	Chain C, Cellular tumor antigen p53 (Homo sapiens)	Homo sapiens	814	814	100%	0.0	100.00%	417	8P1F_C	cellular tumor antigen p53 isoform a (Homo sapiens)	Homo sapiens	813	813	100%	0.0	100.00%	393	NP_000837.3	mutant tumor protein p53 (Homo sapiens)	Homo sapiens	813	813	100%	0.0	99.75%	393	AYE20917.1
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HIT PROTEIN	ORGANISM	QUERY COVER	E- VALUE	IDENTITY	ACCESSION
cellular tumor antigen p53 isoform X1 [Gorilla gorilla gorilla]	Gorilla gorilla gorilla	100%	0.0	99.75%	XP_063556659.1
Chain K, Cellular tumor antigen p53 [Homo sapiens]	Homo sapiens	100%	0.0	100%	7XZZ_K
Chain C, Cellular tumor antigen p53 [Homo sapiens]	Homo sapiens	100%	0.0	100%	8R1F_C
cellular tumor antigen p53 isoform a [Homo sapiens]	Homo sapiens	100%	0.0	100%	NP_000537.3
mutant tumor protein p53 [Homo sapiens]	Homo sapiens	100%	0.0	100%	AYE20617.1

The BLASTP search was performed using the human TP53 protein sequence (UniProt accession P04637) against the NCBI non-redundant (nr) protein database. The query sequence length was **393 amino acids**. The highest-ranking hit was **cellular tumor antigen p53 isoform X1 from Gorilla gorilla gorilla**.

8. SEQUENCE ALIGNMENT :

uniprotkb/P04637

NCBI Blast:sp|P04637|P53_HUMAN

NIH

National Library of Medicine

National Center for Biotechnology Information

BLAST® » blastp suite » results for RID: 9ZH317HX014

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BLAST® » blastp suite » results for RID: 9ZH317HX014

Job Title

sp|P04637|P53_HUMAN Cellular tumor antigen...

RID

9ZH317HX014 Search expires on 09-09 16:16 pm Download All

Program

BLASTP Citation

Database

nr See details

Query ID

lcl|Query_10427660

Description

sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=H...

Molecule type

amino acid

Query Length

393

Other reports

Distance tree of resultsMultiple alignmentMSA viewer

Filter Results

Organism

only top 20 will appear

Type common name, binomial, taxid or group name

Add organism

Percent Identity

E value

Query Coverage

to

to

to

Filter

Reset

Compare these results against the new Clustered nr database

BLAST

Descriptions

Graphic Summary

Alignments

Taxonomy

Alignment view

Pairwise

Restore defaults

Download

100 sequences selected

Download

GenPept

Graphics

cellular tumor antigen p53 isoform X1 [Gorilla gorilla gorilla]

Sequence ID: XP_063556659.1 Length: 408 Number of Matches: 1

Range 1: 16 to 408

GenPept

Graphics

Next Match

Previous Match

Score

Expect

Method

Identities

Positives

Gaps

814 bits(2103) 0.0 Compositional matrix adjust. 392/393(99%) 392/393(99%) 0/393(0%)

Query

1

MEEPQSDPSVEPRLSQETFSDLKLLPENNVLSPLPSQAMDLLSLPDDIEQWFTEDPGP

60

Sbjct

16

MEEPQSDPSVEPRLSQETFSDLKLLPENNVLSPLPSQAMDLLSLPDDIEQWFTEDPGP

75

Query

61

DEAPRMPEAAPVAPAPAAPTPAAPAPAPSNPLSSVPSQKTYQGSYGRFLGFLHSGTAK

120

Sbjct

76

DEAPRMPEAAPVAPAPAAPTPAAPAPAPSNPLSSVPSQKTYQGSYGRFLGFLHSGTAK

135

Query

121

SVTCYSPALNKHFCQLAKTCPVQLWVDSPPPGTRVRAMAIYKQSQHTEVVRRCPEHE

180

Sbjct

136

SVTCYSPALNKHFCQLAKTCPVQLWVDSPPPGTRVRAMAIYKQSQHTEVVRRCPEHE

195

Query

181

RCSDSDGLAPQHILIRVEGNLRVYLDORNTFRHSVAVVYEPPEVSGDCTTTHYNYVCS

240

Sbjct

196

RCSDSDGLAPQHILIRVEGNLRVYLDORNTFRHSVAVVYEPPEVSGDCTTTHYNYVCS

255

Query

241

SCPGQNRNPRLTITITLEDSSGNLLGRNSFEVRVCACPGDRDRTEENLRKKGEPIELP

300

Sbjct

256

SCPGQNRNPRLTITITLEDSSGNLLGRNSFEVRVCACPGDRDRTEENLRKKGEPIELP

315

Query

301

PGSTRALPNTTSSSPQKKKPLDGEVFTLQIRGRERFERFRELNEALELKDAGAKEPG

360

Sbjct

316

PGSTRALPNTTSSSPQKKKPLDGEVFTLQIRGRERFERFRELNEALELKDAGAKEPG

375

Query

361

GSRAHSHLKSCKGQSTSRHKKLRFKTEGPDSD

393

Sbjct

376

GSRAHSHLKSCKGQSTSRHKKLRFKTEGPDSD

408

Download

GenPept

Graphics

Chain K, Cellular tumor antigen p53 [Homo sapiens]

Sequence ID: 7XZZ_K Length: 395 Number of Matches: 1

See 3 more title(s)

See all identical proteins (IPG)

Range 1: 1 to 395

GenPept

Graphics

Next Match

Previous Match

Score

Expect

Method

Identities

Positives

Gaps

814 bits(2102) 0.0 Compositional matrix adjust. 393/393(100%) 393/393(100%) 0/393(0%)

Query

1

MEEPQSDPSVEPRLSQETFSDLKLLPENNVLSPLPSQAMDLLSLPDDIEQWFTEDPGP

60

Sbjct

1

MEEPQSDPSVEPRLSQETFSDLKLLPENNVLSPLPSQAMDLLSLPDDIEQWFTEDPGP

60

Related Information

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

Related Information

Identical Proteins - identical proteins to 7XZZ_K

9. INTERPRETATION :

The BLASTP analysis demonstrated that the human TP53 protein shares a very high degree of sequence similarity with homologous proteins from other organisms. The highest-ranking hit was the cellular tumour antigen p53 isoform X1 from *Gorilla gorilla* . It showed **100% query coverage, 99.75% percentage identity, and an E-value of 0.0**. These results indicate a highly significant and strongly conserved sequence relationship between human TP53 and the identified homolog.

The presence of highly conserved amino acid positions indicates that important portions of the TP53 protein sequence have been maintained during evolution. Such conservation may be associated with important structural and functional roles of the protein. The small number of sequence differences observed between the human and gorilla TP53 proteins represents minor evolutionary variation. Overall, the BLASTP results strongly support that TP53 is a highly conserved protein across closely related species.

10. CONCLUSION :

In this study, the human TP53 protein sequence was successfully retrieved from UniProt and analyzed using NCBI BLASTP. The search identified multiple homologous protein sequences with significant sequence similarity. Analysis of percentage identity, query coverage and E-values supported the presence of conserved regions among the identified proteins. Overall, BLAST analysis provided useful evidence for studying sequence similarity and evolutionary relationships of the TP53 protein.

11. REFERENCES :

1. UniProt Consortium. UniProt: the Universal Protein Knowledgebase.
2. National Center for Biotechnology Information (NCBI). BLAST: Basic Local Alignment Search Tool.
3. Altschul SF, et al. Basic local alignment search tool. *Journal of Molecular Biology*. 1990.